

GOVERN.AI

ANTHERA Systems

OPERATIONAL MANUAL

VERSION

MVP v3.1

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Complete manual - user, implementation, deploy.

Complete operational guide: platform usage, technical extension, production deployment.

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Audience: Compliance Manager, DPO, CISO, AI Engineer, DevOps, Solution Architect

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PART 1 — USER GUIDE

1. Platform introduction

GOVERN.AI is the sovereign control plane for governing AI agents in enterprises operating in regulated industries (banking, insurance, healthcare, public administration, critical infrastructure).

It allows you to:

- **Register** every AI agent with risk classification (EU AI Act)
- **Define** granular policies per regulation
- **Detect** policy conflicts with operational guidance
- **Quantify** compliance posture with a deterministic score (0-100)
- **Guide** Compliance Officers through SOX 404 wizards
- **Trace** every AI action in an explainable audit trail
- **Monitor** real-time 8 international standards
- **Export** PDF/CSV reports ready for regulators and audits

Target personas

Persona	Main need
Compliance Manager / DPO	Unified visibility, quantitative score, gap management
CISO	Risk classification, audit trail, security headers
Auditor (internal/external)	Audit Readiness Score, PDF export, evidence
AI Engineer	Agent registry, lifecycle, policy enforcement
Board / Investor	Intelligence Center, posture score, trend

2. Login, access and security

Access

- Open browser at `http://localhost:3000` (or your deploy URL)
- You will be directed to the **Landing Page**
- Click **"Get Started"** or **"Login"**
- Enter credentials

Demo credentials

- **Username:** admin
- **Password:** AdminGovern2026!

In production change these credentials immediately. See section 21 — Security.

Roles (RBAC)

Role	Read	Write	Delete	Admin	Export
Admin	Yes	Yes	Yes	Yes	Yes
DPO	Yes	Yes	No	No	Yes
Auditor	Yes	No	No	No	Yes
Viewer	Yes	No	No	No	No

Session

- JWT signed HS256, expires after **8 hours**
- Token stored in browser localStorage
- Logout: button at top right of sidebar

3. Main navigation

Sidebar (left)

After login the app shows the dashboard layout with collapsible sidebar and 8 main entries:

Entry	Page	Content
Overview	<code>`/dashboard/overview`</code>	KPI + 3 main charts
Agents	<code>`/dashboard/agents`</code>	AI Agent registry
Policies	<code>`/dashboard/policies`</code>	Policies + Conflict Engine
SOX 404	<code>`/dashboard/sox`</code>	SOX controls wizard
Audit	<code>`/dashboard/audit`</code>	Complete Audit Trail
Compliance	<code>`/dashboard/compliance`</code>	8 regulatory standards
Intelligence	<code>`/dashboard/intelligence`</code>	Score Center (v3.0)
ARIA	<code>`/dashboard/assistant`</code>	AI Assistant

Language switch

Top right: **IT / EN** toggle.

4. Overview Dashboard

The Overview page displays:

- **4 KPI cards:** Total agents, Active policies, Total audit logs, Overall Compliance Score
- **3 Recharts:**
 - Risk distribution (pie chart) - Audit outcome trend (line chart) - Compliance progress per standard (bar chart)
- **Global governance score banner** (v3.0): tile with score, trend, quick link to Intelligence Center

How to read the Compliance Score

- **0-49:** Critical (red) — immediate action
- **50-69:** Warning (yellow) — gaps to close
- **70-89:** Good (light green) — solid posture
- **90-100:** Excellent (dark green) — best-in-class

5. AI Agent Registry

Content

List of all AI agents registered in the organization, with:

Field	Description
Name	Agent name
Type	Type (chatbot, classifier, advisor, monitor, etc.)
Risk Level	low / medium / high / unacceptable (EU AI Act)
Status	active / inactive / under_review
Owner	Responsible team or person
Description	Business function

Operations

- **Create** agent: button "+ Add Agent" -> form dialog
- **Edit:** "pencil" icon on row
- **Delete:** "trash" icon on row (Admin only)
- **Filter:** search bar + status and risk_level filters

Best practices

- Register the agent **before** operational deploy
- Classify risk consistently with EU AI Act

- Associate the agent with one or more policies (see section 6)
- Use descriptive names (e.g., "Credit Scoring AI", "AML Monitor")

6. Policy Engine + Policy Guidance

Policy

A **policy** defines a governance rule applied to one or more agents.

Field	Description
Name	Policy name
Regulation	Reference standard (GDPR, EU AI Act, SOX, ...)
Severity	low / medium / high / critical
Action	allow / log / require_review / block
Description	What the policy does
Status	active / draft / archived

Practical examples

- ***GDPR Policy***: "Log all credit decisions with reasoning" — action=log
- ***EU AI Act Policy***: "Block if biased input detected" — action=block
- ***SOX Policy***: "Require human approval for financial decisions" — action=require_review

Policy Conflict Engine (Step E4)

Clicking **"Run Conflict Scan"** the system automatically detects 4 anomaly types:

Type	Description
Action Conflict	Two policies with conflicting actions on the same agent
Gap	Regulatory standard not covered by any policy
Overlap	Two policies overlapping on the same scope
Redundancy	Duplicate policies with the same action

Policy Guidance Engine (Step E5 — v2.5)

For each detected conflict, GOVERN.AI provides:

- **Contextual guidance**: what to do to resolve it
- **Impact analysis**: priority, urgency, business risk
- **Resolve dialog**: form to record the decision taken

- **Mandatory resolution notes** (auditable)
- **Automatic audit log** of the action

Typical workflow

- Create/import policies for each relevant standard
- Run conflict scan
- Open each conflict -> read guidance -> decide action
- Fill resolution notes -> confirm
- Verify recording in Audit Trail

7. SOX Section 404 Wizard

What it does

Guided wizard for assessing the **20 SOX 404 controls** divided into **5 domains**:

- **Access Control** (4 controls)
- **Change Management** (4 controls)
- **IT Operations** (4 controls)
- **Data Integrity** (4 controls)
- **Security** (4 controls)

Workflow

- Open the **SOX 404** page
- Select a control: initial status **not_started**
- Change status to:
 - **in_progress** (work ongoing) - **completed** (control satisfied) - **failed** (open gap)
- Add evidence/notes in the edit dialog
- Export the **SOX 404 Report PDF**

Audit Readiness Score

At the bottom of the page, a **risk-weighted score** (0-100%) indicates how ready you are for a SOX audit. Computed considering:

- Number of completed vs. total controls
- Domain weight (Security/Access Control = higher weight)
- Number of failed controls (penalty)

8. Compliance Monitor (8 standards)

Tracked standards

Standard	Description	Demo progress
EU AI Act	European AI Regulation	45%
GDPR	Personal Data Protection	78%
ISO 27001	Information Security Mgmt	92%
ISO 42001	AI Management System	34%
DORA	Digital Operational Resilience	61%
NIS2	Cybersecurity Directive	83%
SOX	Sarbanes-Oxley (Section 404)	56%
D.Lgs. 262/2005	Italian Financial Controls	48%

Available actions

- See progress, status, last update per standard
- Filter by status (Compliant / In Progress / Non Compliant)
- Open detail with specific KPIs
- Export **Compliance Report PDF**

9. Audit Trail

Content

Complete log of every traceable platform action:

Field	Description
Timestamp	Date/time ISO 8601 (UTC)
Agent	AI agent involved
Action	Attempted action
Outcome	success / blocked / flagged
Risk Level	low / medium / high
Details	Explainability (JSON)

Filters

- By date (range picker)
- By agent
- By outcome
- By risk level

Export

- **PDF**: report formatted for external audit
- **CSV**: for BI analysis

10. Intelligence Center (v3.0)

Content

Investor-ready premium dashboard with:

- Central **Score Ring** (overall governance score)
- **KPI breakdown**: 4 evaluation dimensions (Agents, Policies, Audit, Compliance)
- **Top/Bottom agents** by score
- **Standard scores** (8 mini-tiles)
- **Positive and negative drivers**: what contributes in pluses/minuses
- **Methodology note**: how the score is computed
- **Insights, risks, remediation**: recommended actions
- **Score History**: snapshot with delta and trend

How the score is computed

Deterministic algorithm (see section 17) considering:

- Agent risk distribution
- Policy status and open conflicts
- Audit outcome trend (success vs blocked)
- SOX 404 controls progress
- 8 standards coverage

Investor use case

The Intelligence Center is designed to be shown in board meetings / due diligence:

- 1 synthetic score (main KPI)
- Historical trend (validation that posture improves)
- Driver list (transparency)
- Remediation (future visibility)

11. ARIA — AI Assistant

What it does

ARIA is an AI advisor expert in the 8 regulations tracked by the platform. It answers in chat with **SSE streaming** with specific knowledge of:

- EU AI Act
- GDPR
- ISO 27001 / 42001
- DORA / NIS2
- SOX / D.Lgs. 262

How to use it

- Open **ARIA** page
- Write a question (e.g., **""What SOX controls should I apply to credit scoring?""**)
- ARIA replies in streaming
- Conversations are **persisted per session**

v3.0 contextualization

ARIA knows in real-time:

- The current governance score
- Open policy conflicts
- SOX control gaps

So it can provide **contextualized** advice on your compliance status.

12. PDF/CSV Export

Report	Endpoint	Format
Audit Trail	<code>`/api/audit/export/pdf`</code> or <code>`/csv`</code>	PDF / CSV
Compliance Report	<code>`/api/compliance/export/pdf`</code>	PDF
SOX 404 Report	<code>`/api/sox/report/pdf`</code>	PDF
Investor Score	from Intelligence Center	PDF (manual)

All reports are generated with **ReportLab** in GOVERN.AI branded style.

13. Internationalization

- **IT/EN** fully translated (~140 i18n keys)

- Top-right toggle
 - All pages, forms, tooltips and toasts are localized
-

PART 2 — IMPLEMENTATION GUIDE

14. Technical architecture

[illegible]

Layers

- **Frontend SPA:** React 19, Tailwind, Shadcn UI, Recharts, custom i18n
- **Modular backend:** server.py registers 10 routers (auth, agents, policies, audit, compliance, dashboard, chat, sox_wizard, policy_engine, score)
- **Service layer:** services/compliance_engine.py for scoring logic
- **Database:** MongoDB with Motor async, 15+ optimized indices
- **LLM:** portable via litellm (default openai/gpt-4o)
- **Security:** JWT HS256, bcrypt, SlowAPI rate limit, 5 security headers, restrictive CORS

15. Data models (MongoDB)

Collections

Collection	Typical docs	Key indices
`users`	1+	id (unique), username (unique)
`agents`	14 demo	id (unique), status, risk_level
`policies`	20+ demo	id (unique), regulation, severity
`audit_logs`	150+ demo	id (unique), timestamp, agent_name, outcome, risk_level
`compliance_standards`	8 demo	id (unique), code (unique)
`chat_messages`	variable	session_id + timestamp (compound)
`sox_controls`	20	id (unique), domain, status

Example: Agent

```
{
  "id": "uuid-1234",
  "name": "Credit Scoring AI",
  "type": "classifier",
  "risk_level": "high",
  "status": "active",
  "owner": "Risk Management",
  "description": "Automated credit decision support",
  "created_at": "2026-05-14T08:00:00Z"
}
```

Example: Policy (with conflict resolution v2.5)

```
{
  "id": "uuid-5678",
  "name": "GDPR – Log all decisions",
  "regulation": "GDPR",
  "severity": "high",
  "action": "log",
  "description": "...",
  "status": "active",
  "conflicts": [{
    "type": "overlap",
    "with_policy_id": "uuid-other",
    "guidance": "Merge policies into a single comprehensive rule",
    "impact": "medium",
    "resolution_notes": "Merged 2026-05-10 by user@example.com",
    "resolved_at": "2026-05-10T10:00:00Z"
  }]
}
```

16. REST API (complete endpoints)

All endpoints are prefixed with `/api`. Auth via `Authorization: Bearer` header.

Authentication

Method	Endpoint	Description
POST	<code>/api/auth/login`</code>	Login (username + password)
POST	<code>/api/auth/register`</code>	Registration (admin only)
GET	<code>/api/auth/me`</code>	Current user profile

Agents

Method	Endpoint	Description
GET	<code>/api/agents`</code>	List agents (with filters)
POST	<code>/api/agents`</code>	Create agent
PATCH	<code>/api/agents/{id}`</code>	Update agent
DELETE	<code>/api/agents/{id}`</code>	Delete agent

Policies & Policy Engine

Method	Endpoint	Description
GET	<code>/api/policies`</code>	List policies
POST	<code>/api/policies`</code>	Create policy
PATCH	<code>/api/policies/{id}`</code>	Update
DELETE	<code>/api/policies/{id}`</code>	Delete
GET	<code>/api/policy-engine/conflicts`</code>	Scan conflicts
GET	<code>/api/policy-engine/conflicts/{id}/guidance`</code>	Detailed guidance (v2.5)
POST	<code>/api/policy-engine/conflicts/{id}/resolve`</code>	Resolve conflict with notes (v2.5)
GET	<code>/api/policy-engine/scan-history`</code>	Scan history

Audit Trail

Method	Endpoint	Description
GET	<code>`/api/audit`</code>	Filtered log list (paginated)
GET	<code>`/api/audit/export/pdf`</code>	PDF export
GET	<code>`/api/audit/export/csv`</code>	CSV export

Compliance

Method	Endpoint	Description
GET	<code>`/api/compliance`</code>	List standards
GET	<code>`/api/compliance/{code}`</code>	Standard detail
GET	<code>`/api/compliance/export/pdf`</code>	PDF report

Dashboard

Method	Endpoint	Description
GET	<code>`/api/dashboard/stats`</code>	Overview KPI

SOX Wizard

Method	Endpoint	Description
GET	<code>`/api/sox/controls`</code>	List 20 controls
PATCH	<code>`/api/sox/controls/{id}`</code>	Update control status
GET	<code>`/api/sox/report`</code>	JSON report
GET	<code>`/api/sox/report/pdf`</code>	PDF report
GET	<code>`/api/sox/readiness-score`</code>	Audit Readiness Score

Compliance Intelligence Engine (v3.0)

Method	Endpoint	Description
GET	<code>/api/score/overview`</code>	Governance score + explainability
GET	<code>/api/score/agents`</code>	Per-agent score
GET	<code>/api/score/agents/{id}`</code>	Single agent score
GET	<code>/api/score/standards`</code>	Per-standard score
GET	<code>/api/score/history`</code>	Snapshot history
GET	<code>/api/score/insights`</code>	Insights, risks, remediation

Chat (ARIA)

Method	Endpoint	Description
POST	<code>/api/chat`</code>	Single message
GET	<code>/api/chat/stream`</code>	SSE streaming
GET	<code>/api/chat/sessions/{id}`</code>	Session history

Interactive documentation

- **Swagger UI:** <http://localhost:8001/docs>
- **ReDoc:** <http://localhost:8001/redoc>

17. Compliance Intelligence Engine

File

- `/app/backend/services/compliance_engine.py`

Scoring logic

The governance score (0-100) is computed as a **weighted average** of:

Dimension	Weight	Base KPI
Agents	0.25	Average risk, active status, policy coverage
Policies	0.25	Active vs draft, open conflicts, standard coverage
Audit	0.20	Success vs blocked outcomes, last-30-day trend
Compliance	0.30	Average progress of 8 standards, critical gaps

Structured output

```
{
  "score": 87,
  "trend": "+4",
  "label": "Good",
  "breakdown": [
    {"dimension": "Agents", "score": 82, "weight": 0.25},
    {"dimension": "Policies", "score": 75, "weight": 0.25},
    {"dimension": "Audit", "score": 91, "weight": 0.20},
    {"dimension": "Compliance", "score": 94, "weight": 0.30}
  ],
  "positive_drivers": ["High ISO 27001 progress", "Low audit-blocked ratio"],
  "negative_drivers": ["EU AI Act gap", "3 open policy conflicts"],
  "methodology": "Weighted multi-dimensional deterministic score..."
}
```

Snapshot history

Each scan creates a snapshot persisted in `score_snapshots`, with timestamp + score + breakdown. This enables **delta/trend** calculation.

18. Policy and standard customization

Adding a standard

- Edit `/app/backend/seed.py` -> STANDARDS array
- Add entry with code, name, description, progress
- Run `python backend/seed.py` to reload

Adding a custom policy

- Via UI: Policies page -> "+ Add Policy"
- Via API: POST `/api/policies` with Pydantic payload

Score weights

To modify weighting, open `/app/backend/services/compliance_engine.py` and change the `DIMENSION_WEIGHTS` dictionary. Weights must sum to 1.0.

19. LLM integration (litellm)

Default

- Provider: OpenAI
- Model: `openai/gpt-4o`
- Variables: `OPENAI_API_KEY`, `LLM_MODEL`

Provider switch

Litellm supports 100+ providers. Examples:

- **Anthropic:** LLM_MODEL=anthropic/claude-3-5-sonnet-20241022 + ANTHROPIC_API_KEY
- **Gemini:** LLM_MODEL=gemini/gemini-1.5-pro + GEMINI_API_KEY
- **Local (Ollama):** LLM_MODEL=ollama/llama3.1 + OLLAMA_API_BASE

Streaming

ARIA uses Server-Sent Events (SSE) via /api/chat/stream. Implemented in routes/chat.py.

20. Frontend extension

Add a page

- Create file in /app/frontend/src/pages/MyPage.js
- Export export default function MyPage() { ... }
- Add route in App.js inside DashboardLayout
- Add sidebar link (DashboardLayout.js)

Add i18n keys

- /app/frontend/src/locales/it.json
- /app/frontend/src/locales/en.json

Use t("my.key") in the component with the LanguageContext hook.

Reusable components

- CrudPage for generic CRUD (see Agents/Policies pages)
- EmptyState for empty states
- SkeletonLoader for loading
- Shadcn components in /components/ui/

21. RBAC and security

Route decorator

```
@router.get("/agents")
@require_role("auditor") # auditor or higher
async def list_agents(...):
    ...
```

Security headers

The `server.py` middleware applies:

- `X-Frame-Options: DENY`
- `X-Content-Type-Options: nosniff`
- `Referrer-Policy: same-origin`
- `Content-Security-Policy: ...`
- `Strict-Transport-Security: ...` (in production)

Rate limiting

SlowAPI applies per-IP limits:

- `/api/auth/login: 5/minute`
- `/api/chat/*: 30/minute`
- Default: 100/minute

Password hashing

bcrypt with cost factor 12. Configured in `routes/auth.py`.

PART 3 — DEPLOY GUIDE

22. Local setup (Docker Compose)

Prerequisites

- Docker >= 24.0
- Docker Compose >= 2.0
- Git

Steps

```
git clone https://github.com/AngeloAng94/GOVERN.AI
cd GOVERN.AI
cp .env.example backend/.env
cp .env.example frontend/.env
docker-compose up --build
```

Open <http://localhost:3000>. Credentials: admin / AdminGovern2026!.

Started services

Service	Port	Description
frontend	3000	React SPA
backend	8001	FastAPI
mongo	27017	MongoDB 7.0

23. Environment variables

Backend (backend/.env)

Variable	Required	Description
`MONGO_URL`	Yes	MongoDB connection
`DB_NAME`	Yes	Database name
`OPENAI_API_KEY`	Yes	LLM API key (or litellm equivalent)
`LLM_MODEL`	No	Default `openai/gpt-4o`
`JWT_SECRET_KEY`	Yes	String >=32 chars
`ALLOWED_ORIGINS`	Yes	CSV of authorized origins

Frontend (frontend/.env)

Variable	Required	Description
<code>REACT_APP_BACKEND_URL`</code>	Yes	Backend public URL

Never commit .env with real values. Use .env.example as template.

24. Initial seed and demo data

The /app/backend/seed.py script loads:

- 1 admin user
- 14 enterprise AI agents
- 20+ policies with realistic conflicts
- 150+ audit logs with 7 incident clusters
- 8 regulatory standards
- 20 SOX 404 controls

Manual execution

```
cd backend
python seed.py
```

The script is idempotent: if data exists, it recreates only if the collection is empty.

25. Production deploy (Docker + Nginx)

Suggested architecture

```
[Cloudflare / WAF]
■ TLS
▼
[Nginx Reverse Proxy]
/ /api
■ ■
[Frontend] [Backend FastAPI]
■
[MongoDB Atlas / Replicaset]
```

Minimal Nginx config

```
server {
  listen 443 ssl http2;
  server_name app.govern.ai;
  ssl_certificate /etc/letsencrypt/live/app.govern.ai/fullchain.pem;
  ssl_certificate_key /etc/letsencrypt/live/app.govern.ai/privkey.pem;

  location /api/ {
    proxy_pass http://backend:8001;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_buffering off; # for SSE
    proxy_read_timeout 3600s;
  }

  location / {
    proxy_pass http://frontend:3000;
  }
}
```

Best practices

- TLS via Let's Encrypt
- WAF (Cloudflare, AWS WAF)
- Managed MongoDB (Atlas)
- Periodic secret rotation for JWT_SECRET_KEY
- Daily backups (see section 27)

26. CI/CD GitHub Actions

The `.github/workflows/ci.yml` workflow runs 4 parallel jobs on every push to `main` or `develop`:

Job	What it does
<code>`backend-tests`</code>	50 pytest against live API + MongoDB
<code>`frontend-build`</code>	<code>`yarn build`</code>
<code>`security-scan`</code>	bandit + safety
<code>`docker-build`</code>	build backend and frontend images

Required secrets

Secret	Required	Description
<code>`OPENAI_API_KEY`</code>	No (skip LLM tests)	For end-to-end LLM tests

27. Monitoring, logging, backup

Logging

- Backend: structured Python log on stdout (captured by Docker)
- Frontend: errors sent to Sentry (optional)

Recommended metrics (Prometheus + Grafana)

- Endpoint latency
- 5xx error rate
- LLM tokens consumed
- Open conflicts count

MongoDB backup

- Atlas: native continuous snapshot
- Self-hosted: daily mongodump cron on S3
- Retention: minimum 30 days

Health check

- GET /api/health -> 200 OK (implement if not already present)
- Docker healthcheck configured in docker-compose.yml

28. Common troubleshooting

Symptom	Probable cause	Solution
Login doesn't work	JWT_SECRET_KEY missing	Set variable and restart
ARIA doesn't respond	Invalid OPENAI_API_KEY	Check key, verify balance
Conflicts not detected	Policy without `regulation`	Fill regulation field
PDF not generated	Missing font in ReportLab	Verify installed fonts
504 backend	Slow MongoDB	Increase `pool_size`, check indices
CORS error	Missing `ALLOWED_ORIGINS`	Add frontend URL
Blank frontend	Wrong `REACT_APP_BACKEND_URL`	Check `.env`

Useful logs

```
docker-compose logs -f backend
docker-compose logs -f frontend
docker-compose logs -f mongo
sudo supervisorctl status # if installed via supervisor
tail -n 100 /var/log/supervisor/backend.*.log
```

29. Appendix — Credentials and useful links

Demo credentials

Role	Username	Password
Admin	`admin`	`AdminGovern2026!`

Change immediately in production.

Links

- **GitHub Repository:** <https://github.com/AngeloAng94/GOVERN.AI>
- **Swagger:** <http://localhost:8001/docs>
- **ReDoc:** <http://localhost:8001/redoc>

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GOVERN.AI — Sovereign Control Plane for Enterprise AI.

A product by ANTHERA Systems.

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